

Ideal gas equation

$$pV = nRT$$

temperature; K

pressure; Pa

vol; m³

amount; mol

constant; 8.31 J K⁻¹ mol⁻¹

Common units conversion:

$$1 \text{ atm} =$$

$$1 \text{ bar} =$$

$$1 \text{ cm}^3 =$$

$$1 \text{ dm}^3 =$$

$$0^\circ\text{C} =$$

Note: "22.7 dm³ mol⁻¹" and "24 dm³ mol⁻¹" can be derived from $pV = nRT$

Ideal gas assumptions

Main assumptions

1. Gas particles have _____ volume
2. Gas particles have _____ intermolecular forces of attraction

Other assumptions

1. Gas particles move in constant random motion
2. Collision between gas particles are perfectly elastic
3. Average kinetic energy of gas particles is proportional to absolute temperature

Real gases

- In a real gas, gas particles have a finite volume and size. There also exists intermolecular forces of attraction.

- Real gases approaches ideality at

- High temperature
- Low pressure

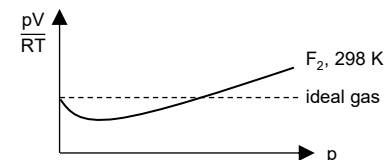
Explanation

high T	
low p	

Deviation from ideal gas

The graphs below shows the behavior of 1 mol of ideal gas and F₂ gas at 298 K

- On the axes below, sketch the graph for 1 mol of Cl₂ at 298 K



- How do the graph of Cl₂ compare to F₂? Explain your answer.

At moderately high pressures

Cl₂ shows _____ negative deviation from ideal gas. This is because Cl₂ has stronger _____ than F₂; Cl₂ has a larger and more polarisable electron cloud, hence _____ id-id. Cl₂ molecules come closer together, and the gas takes up _____ volume.

At very high pressures

Cl₂ shows _____ positive deviation from ideal gas. This is because Cl₂ has a _____ size. As a result, the volume of the gas particles of Cl₂ is _____ significant compared to F₂, and the Cl₂ gas takes up a _____ volume.

Graph sketching

Method

1. Manipulate $pV = nRT$, such that the left side of the equation only contains y-axis variable.
2. Compare the right side of the equation to the x-axis variable; identify constants and deduce if the equation corresponds to:

- $y = k$
- $y = kx$ where k is constant
- $y = kx + c$
- $y = \frac{k}{x}$

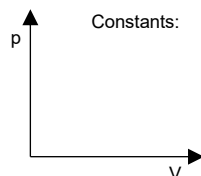
3. Plot graph accordingly

Exceptions:

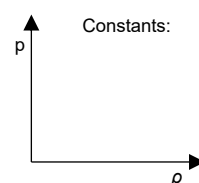
Some equations may not fit those in step 3. If so, plot a graph of x against y , instead of y against x . Then convert the graph back to a y against x .

Plot the following graphs and state the constants

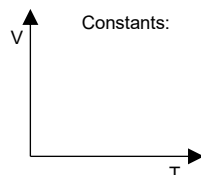
Plot p against V



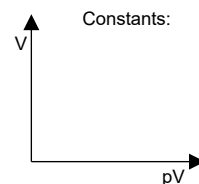
Plot p against density (ρ)



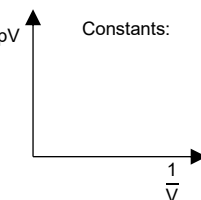
Plot V against T



Plot V against pV



Plot pV against $\frac{1}{V}$



Partial pressure (Dalton's Law)

A sealed 5m³ vessel contains 6 mol of N₂ and 4 mol of O₂ gas at 298 K.

6 mol of N₂
4 mol of O₂

5m³

- Calculate the partial pressure of N₂ in the vessel

Method 1	Method 2

1 mol of Ar gas is added into the same sealed vessel. (assume V & T constant)

- How does the total pressure of the vessel change?
- How does the partial pressure of N₂ in the vessel change?

Ideal gas equation

$$pV = nRT$$

temperature; K

pressure; Pa

vol; m³

amount; mol

constant; 8.31 J K⁻¹ mol⁻¹

Common units conversion:

$$1 \text{ atm} = 101\,325 \text{ Pa}$$

$$1 \text{ bar} = 100\,000 \text{ Pa}$$

$$1 \text{ cm}^3 = 10^{-6} \text{ m}^3$$

$$1 \text{ dm}^3 = 10^{-3} \text{ m}^3$$

$$0^\circ\text{C} = 273 \text{ K}$$

Note: "22.7 dm³ mol⁻¹" and "24 dm³ mol⁻¹" can be derived from $pV = nRT$

Ideal gas assumptions

Main assumptions

- Gas particles have negligible volume
- Gas particles have negligible intermolecular forces of attraction

Other assumptions

- Gas particles move in constant random motion
- Collision between gas particles are perfectly elastic
- Average kinetic energy of gas particles is proportional to absolute temperature

Real gases

- In a real gas, gas particles have a finite volume and size. There also exists intermolecular forces of attraction.

- Real gases approaches ideality at

- High temperature
- Low pressure

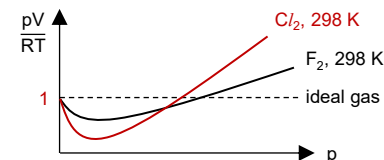
Explanation

high T	gas particles have sufficient kinetic energy to overcome the intermolecular forces of attraction
low p	gas particles are sufficiently far apart and therefore intermolecular forces of attraction is negligible volume occupied by gas particles is negligible compared to container

Deviation from ideal gas

The graphs below shows the behavior of 1 mol of ideal gas and F₂ gas at 298 K

- On the axes below, sketch the graph for 1 mol of Cl₂ at 298 K



- How do the graph of Cl₂ compare to F₂? Explain your answer.

At moderately high pressures

Cl₂ shows greater negative deviation from ideal gas. This is because Cl₂ has stronger intermolecular forces of attraction than F₂; Cl₂ has a larger and more polarisable electron cloud, hence stronger id-id. Cl₂ molecules come closer together, and the gas takes up less volume.

At very high pressures

Cl₂ shows greater positive deviation from ideal gas. This is because Cl₂ has a larger size. As a result, the volume of the gas particles of Cl₂ is more significant compared to F₂, and the Cl₂ gas takes up a larger volume.

Graph sketching

Method

- Manipulate $pV = nRT$, such that the left side of the equation only contains y-axis variable.
- Compare the right side of the equation to the x-axis variable; identify constants and deduce if the equation corresponds to:

- $y = k$
- $y = kx$ where k is constant
- $y = kx + c$
- $y = \frac{k}{x}$

- Plot graph accordingly

Exceptions:

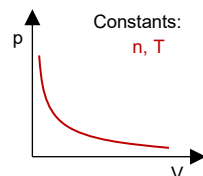
Some equations may not fit those in step 3. If so, plot a graph of x against y , instead of y against x . Then convert the graph back to a y against x .

Plot the following graphs and state the constants

Plot p against V

$$pV = nRT$$

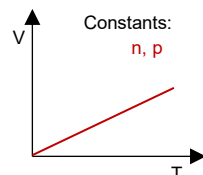
$$p = \frac{nRT}{V}$$



Plot V against T

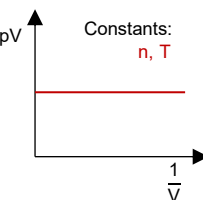
$$pV = nRT$$

$$V = \frac{nRT}{p}$$



Plot pV against $\frac{1}{V}$

$$pV = nRT$$

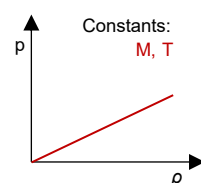


Plot p against density (ρ)

$$pV = nRT$$

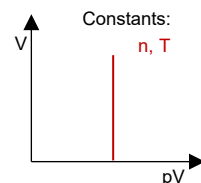
$$pV = \frac{m}{M}RT$$

$$p = \frac{1}{M} \rho RT$$



Plot V against pV

$$pV = nRT$$



Partial pressure (Dalton's Law)

A sealed 5m³ vessel contains 6 mol of N₂ and 4 mol of O₂ gas at 298 K.

- Calculate the partial pressure of N₂ in the vessel

6 mol of N₂
4 mol of O₂
5m³

Method 1	Method 2
$p_{\text{total}}V = n_{\text{total}}RT$ $p_{\text{total}}(5) = (10)(8.31)(298)$ $p_{\text{total}} = 4953 \text{ Pa}$ $p_{\text{N}_2} = \frac{n_{\text{N}_2}}{n_{\text{total}}} p_{\text{total}}$ $= \frac{6}{10} (4953)$ $= 2970 \text{ Pa}$	$p_{\text{N}_2}V = n_{\text{N}_2}RT$ $p_{\text{N}_2}(5) = (6)(8.31)(298)$ $p_{\text{N}_2} = 2970 \text{ Pa}$

1 mol of Ar gas is added into the same sealed vessel. (assume V & T constant)

- How does the total pressure of the vessel change? **increase**
- How does the partial pressure of N₂ in the vessel change? **constant**